



Create your portfolio website in R using Quarto and GitHub Pages

Why?

- Because it is great to share your projects and show future employees what you can do

How?

- **Rstudio** will be the tool we do the work in. You will use new git features in it.
- Using R **Markdown** you will include text, images and links (or even R code!!)
- **Quarto** ('fancy markdown tool') will translate your markdown into HTML website
- **GitHub** Pages will provide free parking ('hosting') for that HTML so people can see it in their browser by going to your portfolio URL



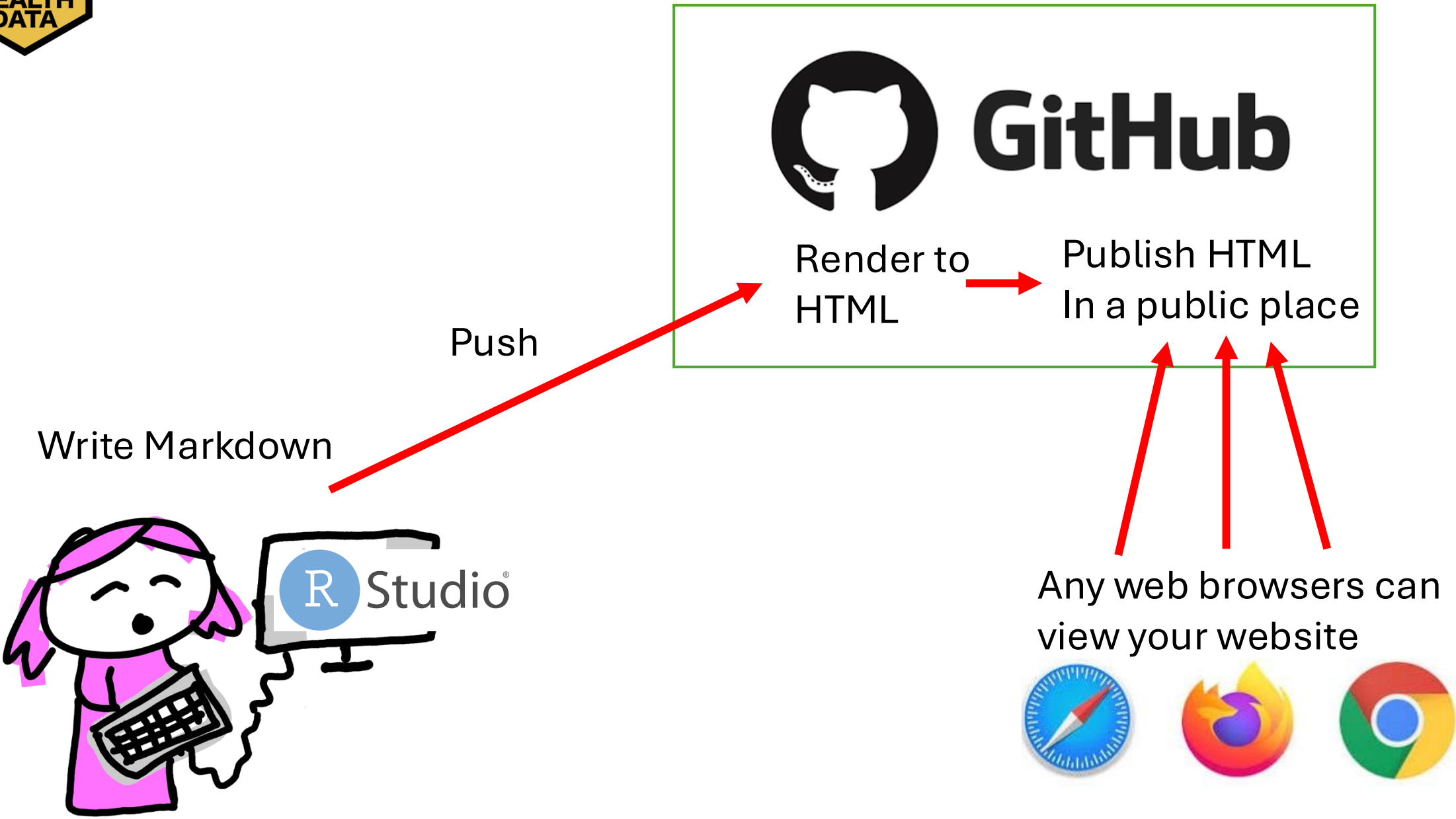
Create your portfolio website in R using Quarto and GitHub Pages

Over a few workshops we will work on:

1. **Creating a simple portfolio with website navigation, available on the Internet**
2. Including blog, projects, images, links and code to showcase your portfolio
3. Customising and tweaking anything you wish
4. Include live R code, graphs and maps



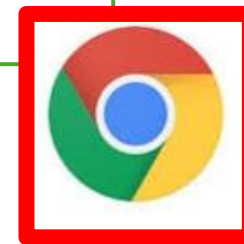
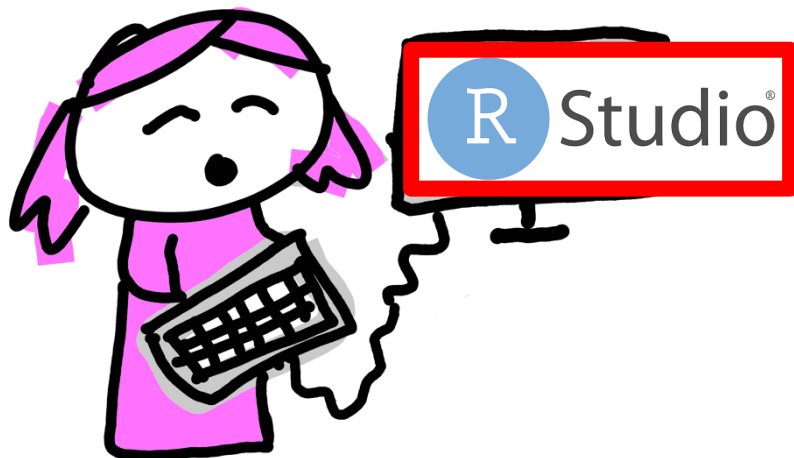
How it all fits together





How it all fits together

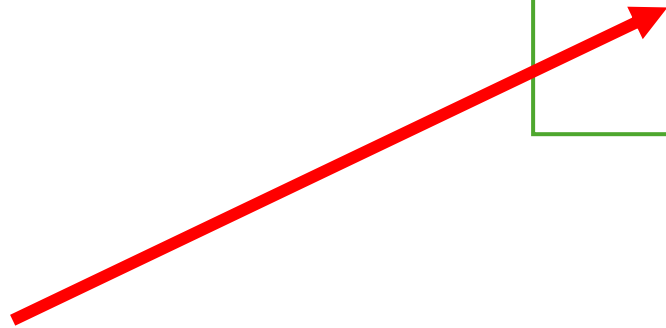
We will work primarily in RStudio. For this workshop we will do it via Noteable server (or your own laptop)



With a few things we'll have to click on the GitHub page via a web browser



Next: connect RStudio with GitHub





Demo: connect RStudio with github

In Rstudio (e.g. on Noteable)

Open your Rstudio > Console

- No need to have a project opened (these are settings applicable to all projects)
- Run below code in console. It will open a website

```
` `` {r}
```

```
# if you don't have 'usethis' library yet, first do install.packages("usethis")
```

```
library(usethis)
```

```
# this will do some cleanup first
```

```
usethis::git_default_branch_configure(name = "main")
```

```
# this will open a website in a browser tab.
```

```
usethis::create_github_token()
```

```
` ``
```

This page will
open
automatically

Give your token a
very meaningful
name, like:

“Rstudio in
Noteable for github
page workshop
August 2026 ”

Settings / Developer Settings

GitHub Apps

OAuth Apps

Personal access tokens

Fine-grained tokens

Tokens (classic)

New personal access token (classic)

Personal access tokens (classic) function like ordinary OAuth access tokens. They can be used instead of a password for Git over HTTPS, or can be used to [authenticate to the API over Basic Authentication](#).

Note

DESCRIBE THE TOKEN'S USE CASE

What's this token for?

Expiration

30 days (Sep 25, 2026)

The token will expire on the selected date

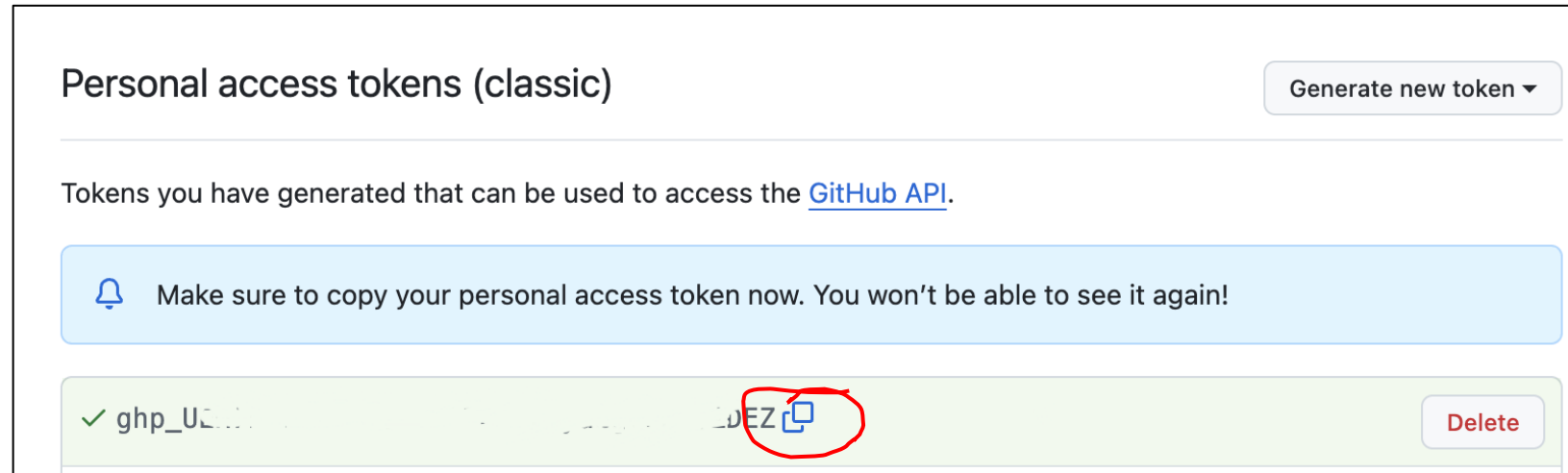
Select scopes

Scopes define the access for personal tokens. [Read more about OAuth scopes.](#)

☒ repo	Full control of private repositories
☒ repo:status	Access commit status
☒ repo_deployment	Access deployment status
☒ public_repo	Access public repositories
☒ repo:invite	Access repository invitations
☒ security_events	Read and write security events
☒ workflow	Update GitHub Action workflows

Generate token Cancel

You will be shown this token only once,
So copy it



Don't close this page yet,
in case copy-pasting doesn't work!

Btw. Never share your token with people, and if you have doubts if it got shared, cancel the token and make a new one



Demo: connect RStudio with github

In Rstudio (e.g. on Noteable)

Back in Rstudio > Console

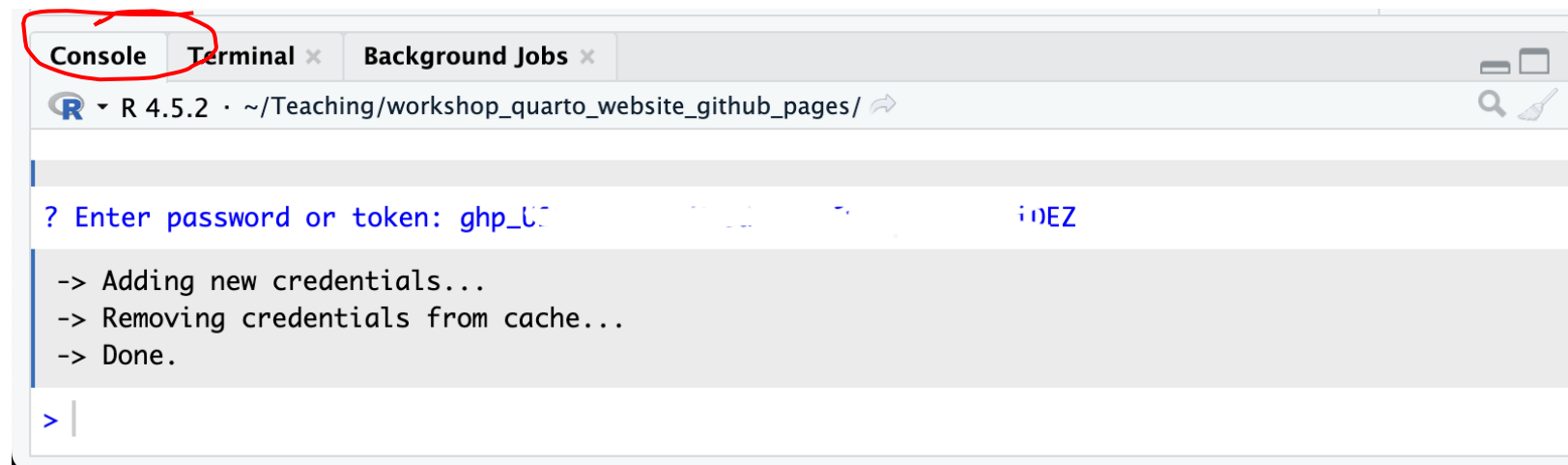
- When asked, paste the token into Console in RStudio

```
` `` {r}
```

if you have an older token, and want to replace it, pick option 2 when asked.

```
gitcreds::gitcreds_set()
```

```
` `` `
```





Demo: connect RStudio with github

In RStudio (e.g. on Noteable)

Back in Rstudio > Console

- let's check if it worked

```
` `` {r}
```

this will either complain, or show your credentials (which means it worked!)

```
gitcreds::gitcreds_get(use_cache = FALSE)
```

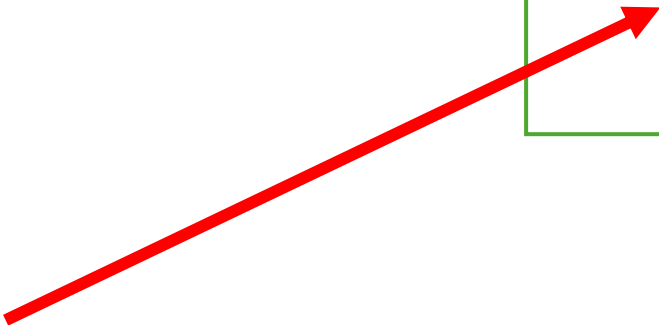
```
` `` `
```

A screenshot of the RStudio Console window. The window has tabs for "Console", "Terminal", and "Background Jobs". The "Console" tab is active. The console shows the R prompt ">" followed by the command "gitcreds::gitcreds_get(use_cache = FALSE)". The output is displayed below the command: "<gitcreds>", "protocol: https", "host : github.com", "username: PersonalAccessToken", and "password: <-- hidden -->". The prompt ">" is followed by a vertical bar "|".

```
Console Terminal Background Jobs
R 4.5.2 · ~/Teaching/workshop_quarto_website_github_pages/
> gitcreds::gitcreds_get(use_cache = FALSE)
<gitcreds>
  protocol: https
    host   : github.com
username: PersonalAccessToken
password: <-- hidden -->
> |
```



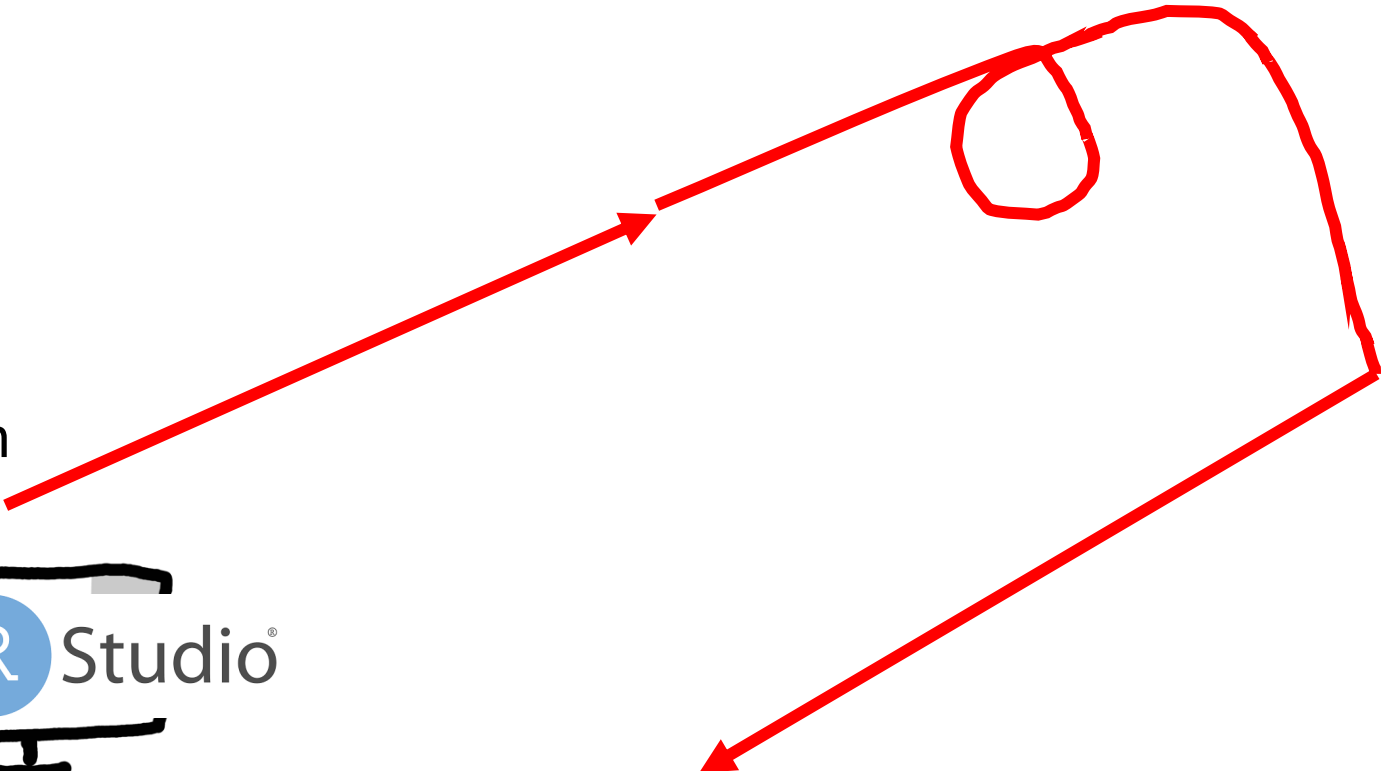
DONE





Next: Write a website in markdown. Preview locally.

Write Markdown





Rendering markdown into HTML (or pdf, or anything visible)



Rendering markdown into HTML (or pdf, or anything visible)

Markdown

this is a header

second header



Rendered HTML, pdf, image

THIS IS A HEADER

Second Header



Rendering markdown into HTML (or pdf, or anything visible)

Markdown

this is a header

second header

- banana
- kiwi

[my blog](<https://myspace.com/banana>)

![banana icon](/images/banana.jpg)



Rendered HTML, pdf, image

THIS IS A HEADER

Second Header



- ◆ banana
- ◆ kiwi

my blog

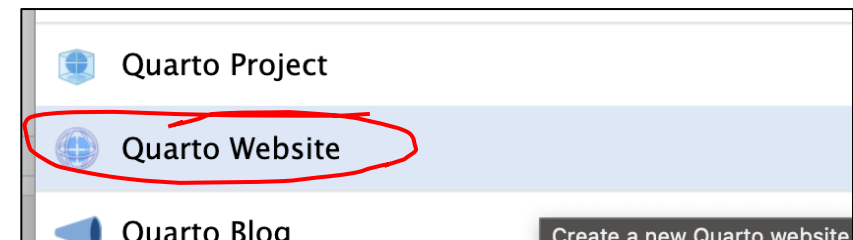
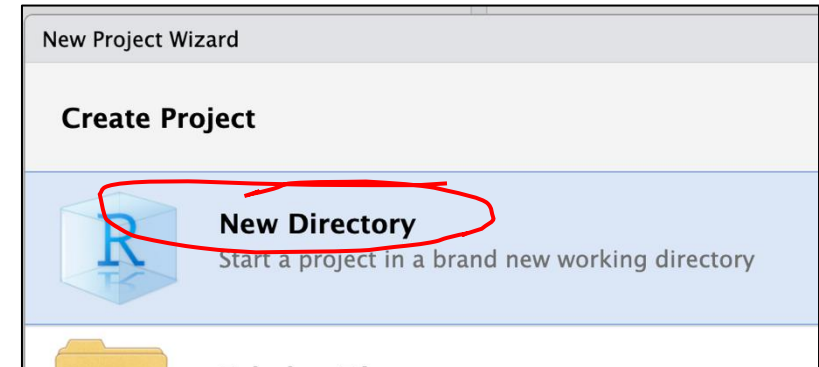
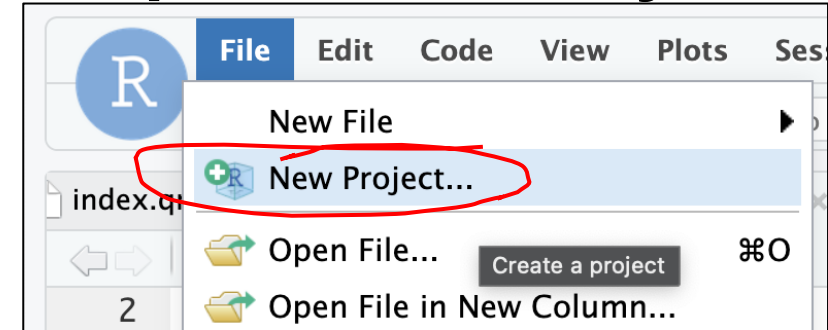




Demo: create Quarto page, edit and preview locally

In RStudio select:

- File > New Project
- New Directory
- Quarto Website





Demo: create Quarto page, edit and preview locally

In RStudio select:

- Give it a name and pick folder
- Select Knitr
- ☒ 'Create git repository'
- ☒ Disable other options
- Click 'Create Project'

The image shows the 'New Project Wizard' dialog box in RStudio. The title bar says 'New Project Wizard'. There is a 'Back' button on the left. The main heading is 'Create Quarto Website'. On the left side of the main area is a blue globe icon. The 'Directory name:' field contains 'workshop_quarto_website_6'. The 'Create project as subdirectory of:' field contains '~' and a 'Browse...' button. The 'Engine:' dropdown is set to 'Knitr'. There are three checkboxes: 'Create a git repository' (checked), 'Use renv with this project' (unchecked), and 'Use visual markdown editor' (unchecked). At the bottom right, there are two buttons: 'Create Project' and 'Cancel'. Red circles highlight the 'Directory name' field, the 'Create a git repository' checkbox, and the 'Create Project' button.



Demo: create Quarto page, edit and preview locally

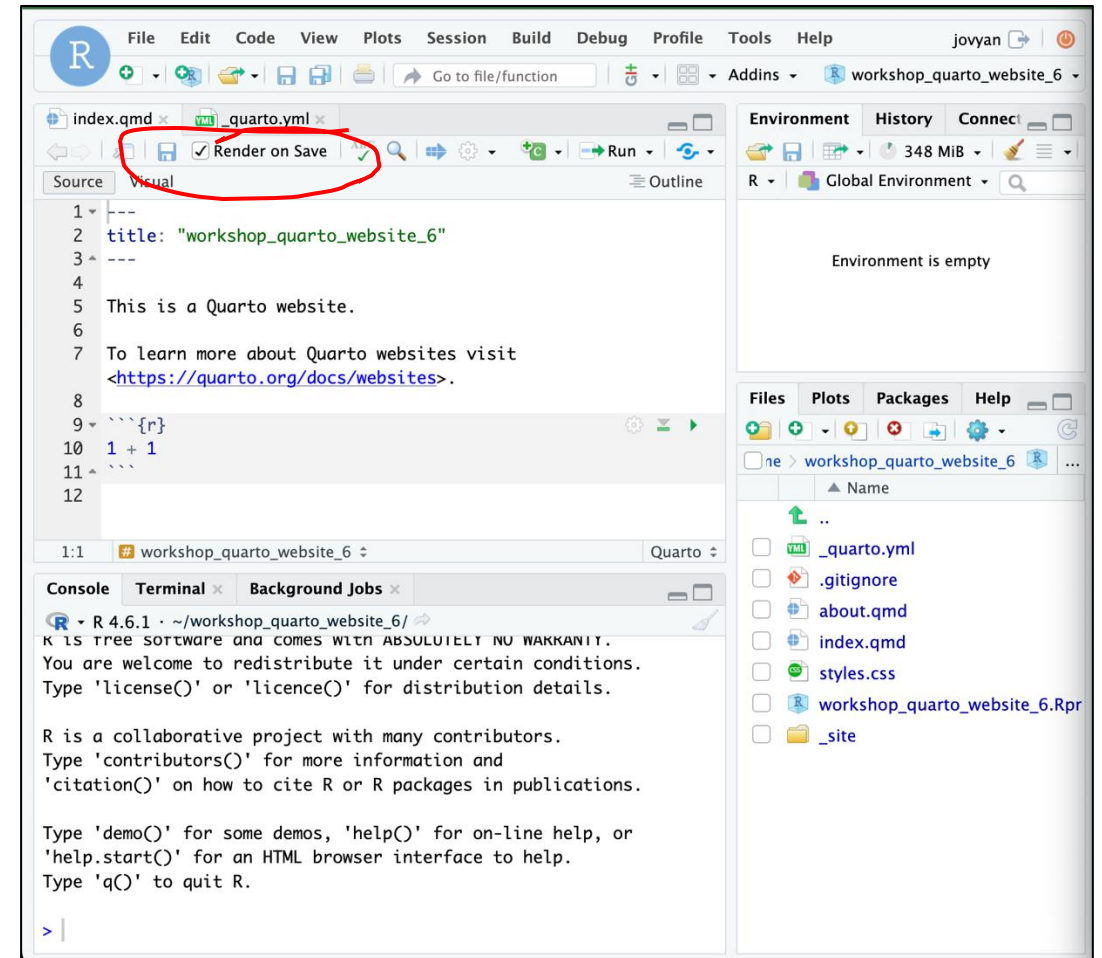
To see a live (local) preview of your website in RStudio:

click Render



Or:

- Tick  'Render on Save'
- make any change, and Save file





Depending on your settings, website preview will show in 'Viewer' panel or in a browser tab:

Render 

Use Visual Editor ⌘F4

Preview in Window

☒ Preview in Viewer Pane

(No Preview)

☒ Preview Images and Equations

☒ Show Previews Inline

☒ Chunk Output Inline

Chunk Output in Console

Expand All Output

Collapse All Output

Clear Output

Clear All Output

Environment History Connections Build Git Tutorial

Files Plots Packages Help Viewer Presentatio

workshop_quarto_...

workshop_quarto_website_6

This is a Quarto website.

To learn more about Quarto websites visit <https://quarto.org/docs/websites>.

1 + 1

[1] 2

workshop_quarto_website_6

noteable.edin...

workshop_quarto_w...

workshop_quarto_website_6

This is a Quarto website.

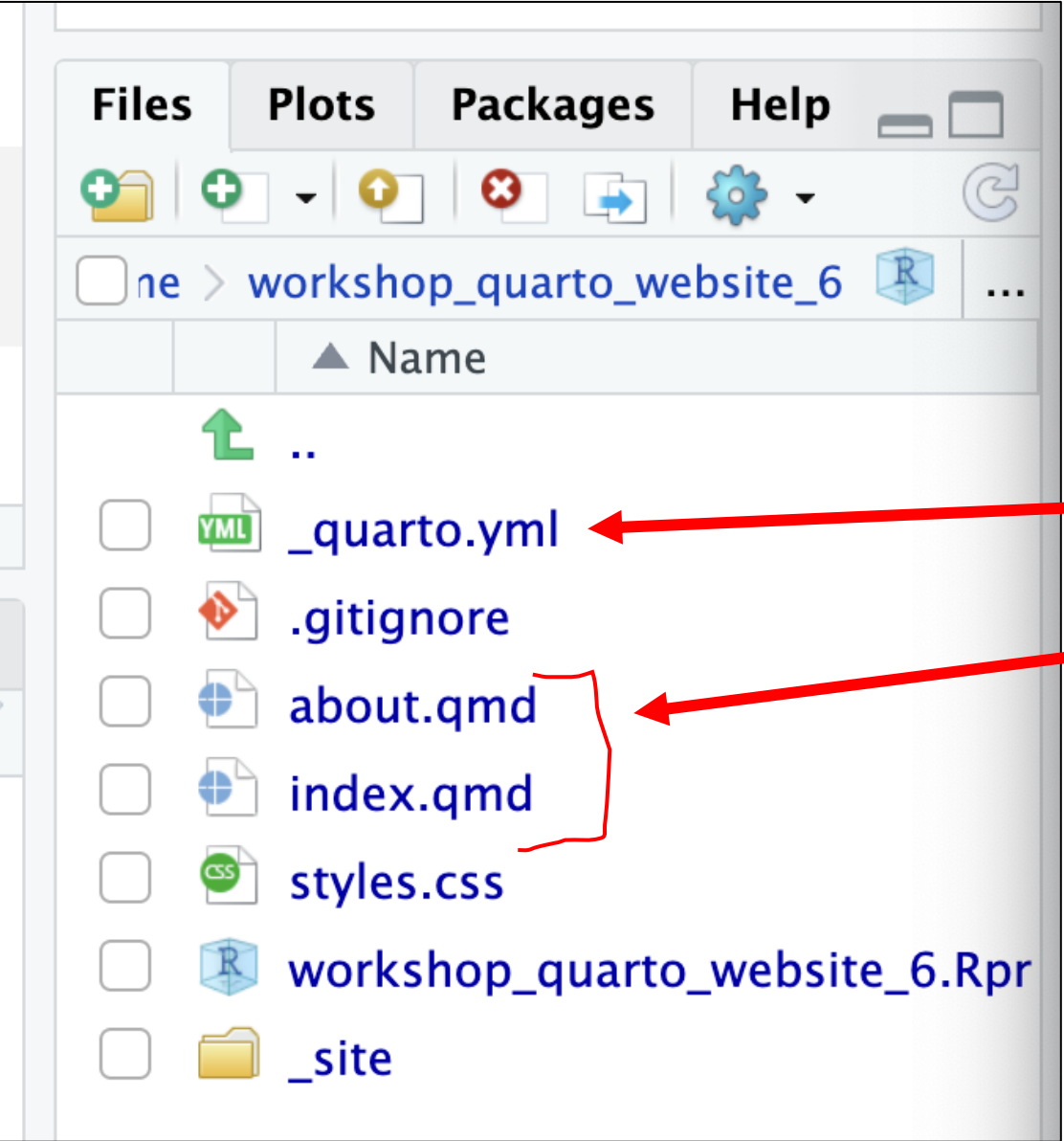
To learn more about Quarto websites visit <https://quarto.org/docs/websites>.

1 + 1

[1] 2



Simple website structure: Pages and Menus



Menu

Pages

👉 index (finger) is for pointing at stuff



Simple website structure: Pages and Menus

Page is a separate markdown file with some content.

- At the top is '**about/meta**' information (between --- tags)
- Below it can contain any **markdown** (images, links, etc).

```
---
title: "workshop_quarto_website_6"
image: profile.jpg
---

This is a Quarto website.

To learn more about Quarto websites visit
<https://quarto.org/docs/websites>.

```{r}
1 + 1
```
```

workshop_quarto_website_6

This is a Quarto website.

To learn more about Quarto websites visit
<https://quarto.org/docs/websites>.

```
1 + 1
```

```
[1] 2
```



Simple website structure: Pages and Menus

Menu (the same everywhere)

- in a separate 'structure' file `_quarto.yml` which defines: Text, Style, Order
- later explore Quarto documentation, it's full of wonderful things you can do!

```
website:  
  title: "workshop_quarto_website_6"  
  navbar:  
    left:  
      - about.qmd  
      - href: index.qmd  
        text: Howdy!  
      - href: hobby.qmd  
        icon: bicycle  
        text: RingRing!
```



worksho...

About

Howdy!

🚲 RingRing!



Rendering (interpreting code) is expensive at scale

A large, green, hand-drawn style starburst or jagged circle shape that frames the text "COMPUTER SCIENCE STUFF:". It has approximately 12 points of varying lengths and angles.

COMPUTER SCIENCE STUFF:



Rendering (interpreting code) is expensive at scale

| | Code / Instructions / Structure | Result |
|--------------|---------------------------------|--------------------------------|
| How it looks | $2+3$ | 5 |
| What is it | text | pictures, tables, text, colour |



Rendering (interpreting code) is expensive at scale

| | Code / Instructions / Structure | Result |
|-----------------|---------------------------------|--------------------------------|
| How it looks | 2+3 | 5 |
| What is it | text | pictures, tables, text, colour |
| Who can read it | expert / coder | anyone |
| Where it lives | in the code file (forever) | where it ran (temporary) |
| Cost to view it | little | little * how_many_times_it_ran |

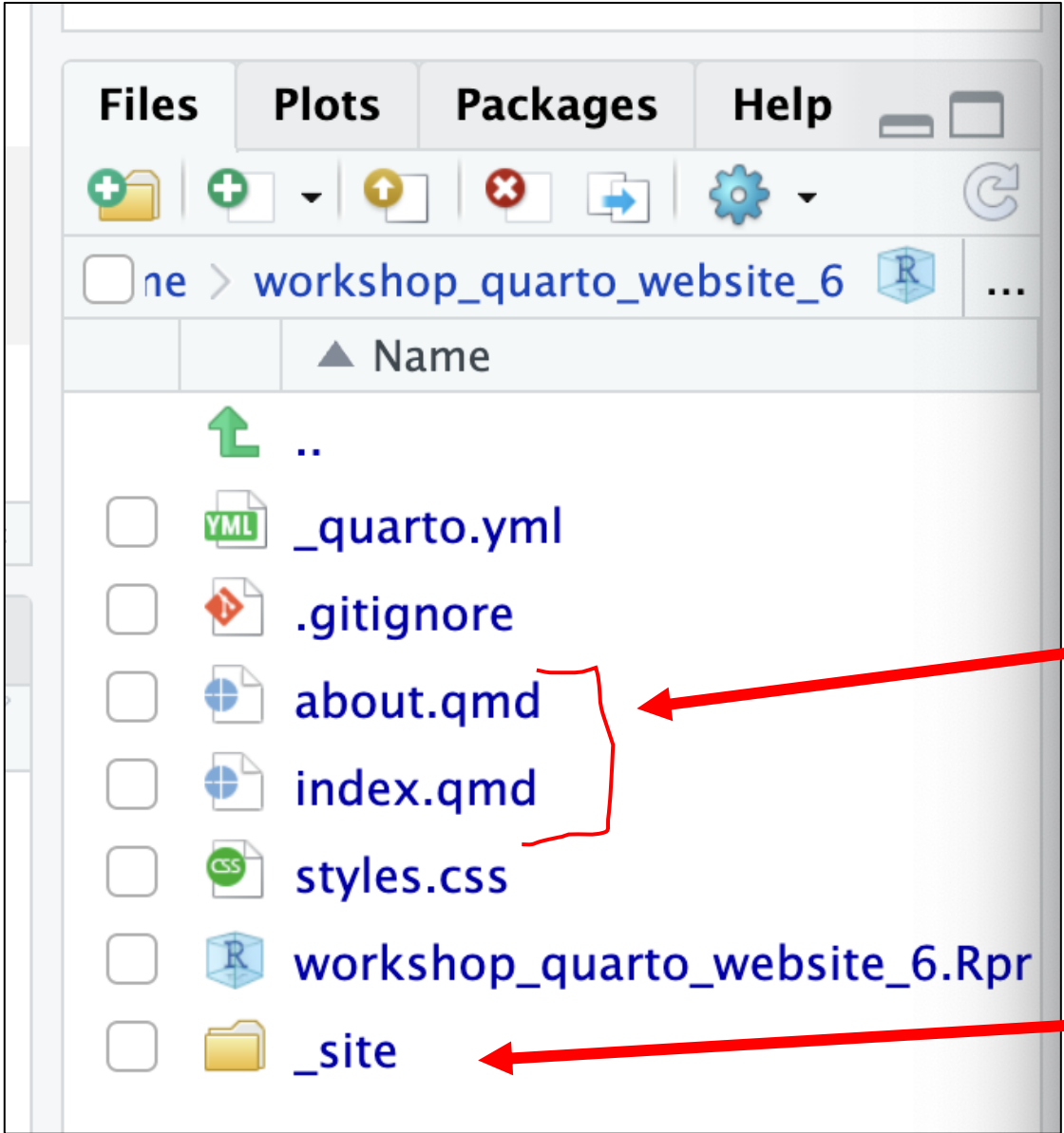


Rendering (interpreting code) is expensive at scale

| Code / Instructions / Structure | | Result |
|--|-------------------------------|--------------------------------|
| How it looks | 2+3 | 5 |
| What is it | text | pictures, tables, text, colour |
| Who can read it | expert / coder | anyone |
| Where it lives | in the code file (forever) | where it ran (temporary) |
| Cost to view it | little | little * how_many_times_it_ran |
| Pre-Render | 2+3 ---> 5 (save as text/img) | 5 (show text) |
| Great or scaling, privacy and security!! | | |



Simple website structure: Pages and Menus



Pages as Markdown (for editing)

Pages rendered as HTML
(for viewing in a web browser)



Next: Tell GitHub how and where to render and publish





Let's get your page online



Let's get your page online

1. Commit your changes to the website

This means you are happy with the changes and want to keep them.

Environment

History

Connections

Build

Git

Diff

Commit

Pull

Push

| Staged | Status | Path |
|-------------------------------------|--------|-------------------------|
| ☒ | A | .gitignore |
| ☒ | A | _quarto.yml |
| ☒ | A | about.qmd |
| ☒ | A | index.qmd |
| ☒ | A | styles.css |
| ☒ | A | workshop_quarto_website |
| ☒ | A | _site/about.html |

Changes

History

(no branch)

Stage

Revert

Ignore

Pull

Push

| Staged | Status | Path |
|-------------------------------------|--------|---------------------------------|
| ☒ | A | .gitignore |
| ☒ | A | _quarto.yml |
| ☒ | A | about.qmd |
| ☒ | A | index.qmd |
| ☒ | A | styles.css |
| ☒ | A | workshop_quarto_website_6.Rproj |
| ☒ | A | _site/about.html |
| ☒ | A | _site/index.html |
| ☒ | A | _site/search.json |
| ☒ | A | site/styles.css |

Commit message

28 characters

first changes to the website

☐ Amend previous commit

Commit

Show

☐ Staged

☒ Unstaged

Context

5 lines

☐ Ignore Whitespace

☒ Stage All

☐ Discard All

Later this will be important



Let's get your page online

2a. Create a remote repository

- In web browser
- On github.com/new create new repo
- give it a meaningful name.
- Keep it public
- do **not** add README or .gitignore

Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).
Required fields are marked with an asterisk (*).

1 General

Owner *



Repository name *

workshop_quarto_website_6

✓ workshop_quarto_website_6 is available.

Great repository names are short and memorable. How about [symmetrical-engine](#)?

Description

0 / 350 characters

2 Configuration

Choose visibility *

Choose who can see and commit to this repository

Public

Start with a template

Templates pre-configure your repository with files.

No template

Add README

READMEs can be used as longer descriptions. [About READMEs](#)

Off

Add .gitignore

.gitignore tells git which files not to track. [About ignoring files](#)

No .gitignore

Add license

Licenses explain how others can use your code. [About licenses](#)

No license

Connect GitHub Apps

Connect this repository to apps drpawelo is subscribed to

Choose apps

Create repository



Let's get your page online

2b. Connect local and remote – get the details oh GitHub website

- Make sure HTTPS is selected
- Copy the bottom section 'push existing repo'

A screenshot of a GitHub repository page for 'workshop_quarto_website_6'. The repository is public. The page shows the 'Quick setup' section with three options: 'Set up in Desktop', 'HTTPS', and 'SSH'. The 'HTTPS' option is circled in red. Below this, there is a text input field containing the repository URL: 'https://github.com/drpawelo/workshop_quarto_website_6.git'. Below the input field, there is a link to 'creating a new file' and a link to 'uploading an existing file'. The 'uploading an existing file' link is also circled in red. Below the 'Quick setup' section, there is a section titled '...or create a new repository on the command line' with a list of git commands. Below this, there is a section titled '...or push an existing repository from the command line' with a list of git commands. This section is circled in red. The repository name 'workshop_quarto_website_6' is at the top left, and the 'Public' label is next to it. At the top right, there are buttons for 'Pin', 'Watch' (0), 'Fork' (0), and 'Star' (0).



Let's get your page online

2c. Connect local and remote – use the details in RStudio

- In RStudio paste those copied lines into the Terminal (NOT CONSOLE)
- It'll connect repos and PUSH.
- Markdown and HTML are online - refresh the github page to see it

A screenshot of the RStudio interface. The 'Terminal' tab is selected and highlighted with a red circle. The terminal shows the following commands and output:

```
(base) jovyan@noteable:~/workshop_quarto_website_6$ git remote add origin https://github.com/dr pawelo/workshop_quarto_website_6.git
git branch -M main
git push -u origin main
Compressing objects: 100% (33/33), done.
Writing objects: 100% (36/36), 353.75 KiB | 9.56 MiB/s, done.
Total 36 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
To https://github.com/dr pawelo/workshop_quarto_website_6.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```



Let's get your page online

2c. Connect local and remote – use the details in RStudio

- In RStudio paste those copied lines into the Terminal. It'll connect repos and PUSH.
- Markdown and HTML are online - refresh the github page to see it

BUT WAIT! HTML is still a bunch of text, instead of colours and pictures :(

A screenshot of the RStudio interface's terminal pane. The 'Terminal' tab is selected and highlighted with a red circle. The terminal shows the following commands and output:

```
(base) jovyan@noteable:~/workshop_quarto_website_6$ git remote add origin https://github.com/dr pawelo/workshop_quarto_website_6.git
git branch -M main
git push -u origin main
Compressing objects: 100% (33/33), done.
Writing objects: 100% (36/36), 353.75 KiB | 9.56 MiB/s, done.
Total 36 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
To https://github.com/dr pawelo/workshop_quarto_website_6.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```



What's on Github?

drpawelo / workshop_quarto_website_6

CodeIssuesPull requestsActionsProjectsWikiMore

Filesmainworkshop_quarto_website_6 / _site

456 small changes and typosf7d9743 · now

| Name | Last commit message | Last commit date |
|-------------|-------------------------|------------------|
| .. | | |
| site_libs | small changes and typos | now |
| about.html | small changes and typos | now |
| index.html | small changes and typos | now |
| search.json | d | 14 hours ago |
| styles.css | d | 14 hours ago |

drpawelo / workshop_quarto_website_6

<> CodeIssuesPull requestsActionsProjectsWiki

mainworkshop_quarto_website_6 / _site / about.html

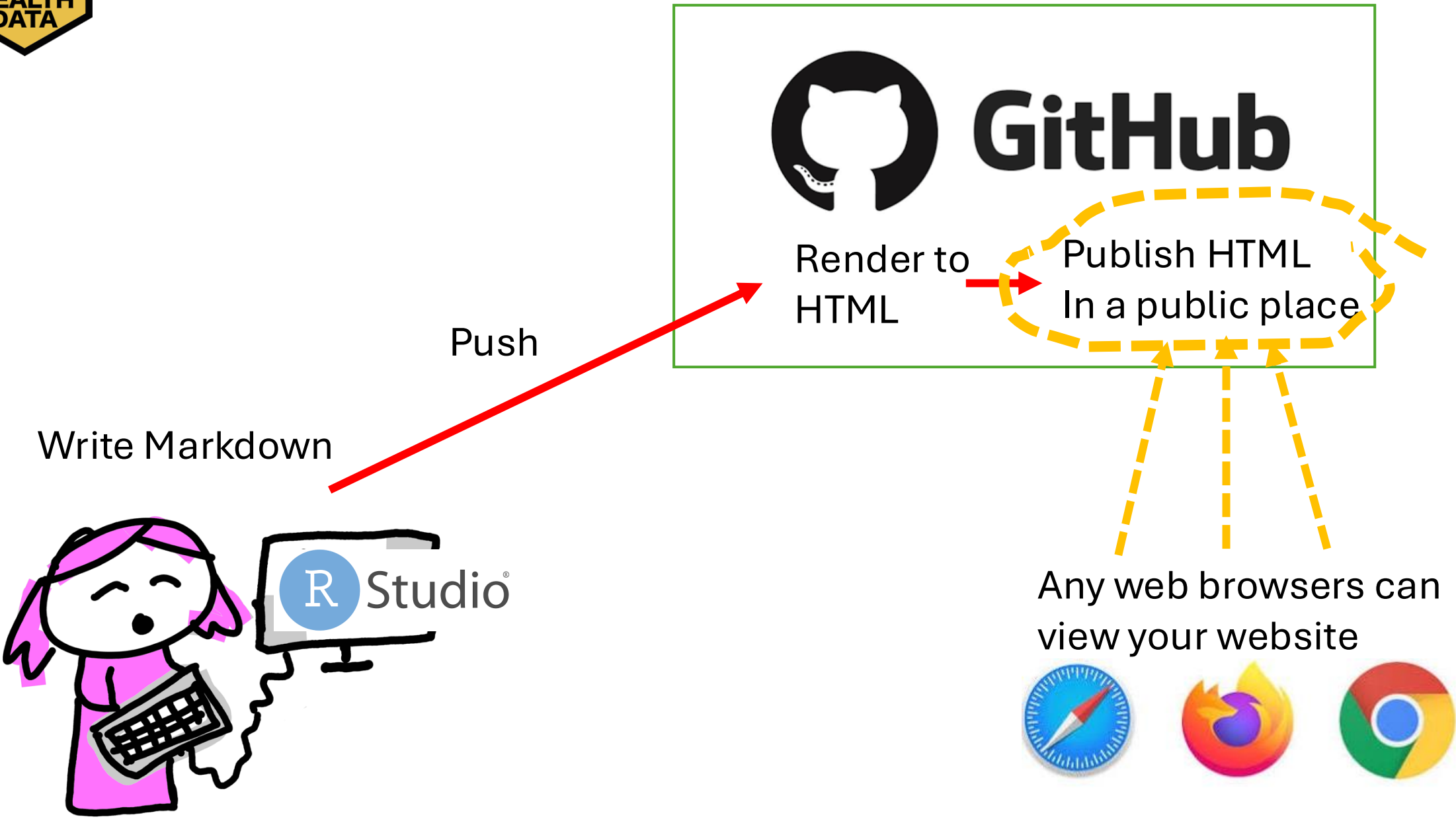
456 d

CodeBlame586 lines (555 loc) · 21.6 KB

```
1 <!DOCTYPE html>
2 <html xmlns="http://www.w3.org/1999/xhtml" lang="en" xml:lang="en"
3
4 <meta charset="utf-8">
5 <meta name="generator" content="quarto-1.7.32">
6
7 <meta name="viewport" content="width=device-width, initial-scale=1
8
9
10 <title>About – workshop_quarto_website_6</title>
11 <style>
12 code{white-space: pre-wrap;}
13 span.smallcaps{font-variant: small-caps;}
14 div.columns{display: flex; gap: min(4vw, 1.5em);}
15 div.column{flex: auto; overflow-x: auto;}
16 div.hanging-indent{margin-left: 1.5em; text-indent: -1.5em;}
17 ul.task-list{list-style: none;}
18 ul.task-list li input[type="checkbox"] {
19   width: 0.8em;
20   margin: 0 0.8em 0.2em -1em; /* quarto-specific, see https://gith
21   vertical-align: middle;
```



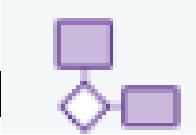
How it all fits together

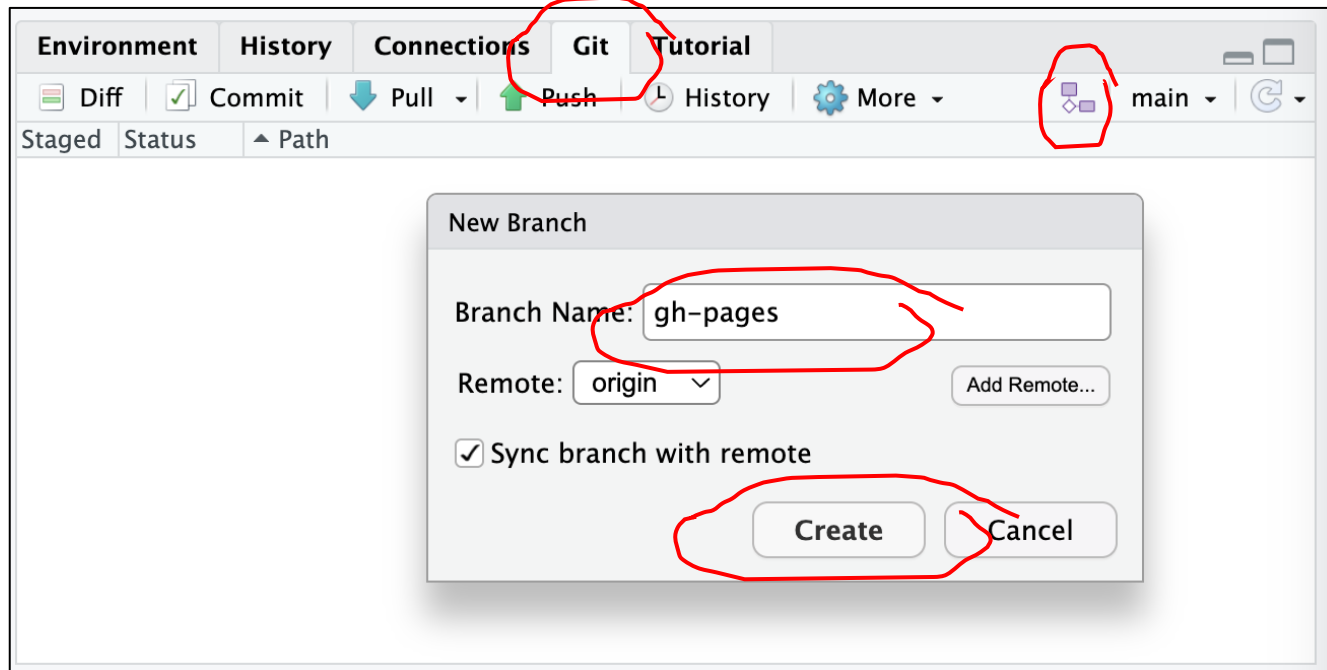




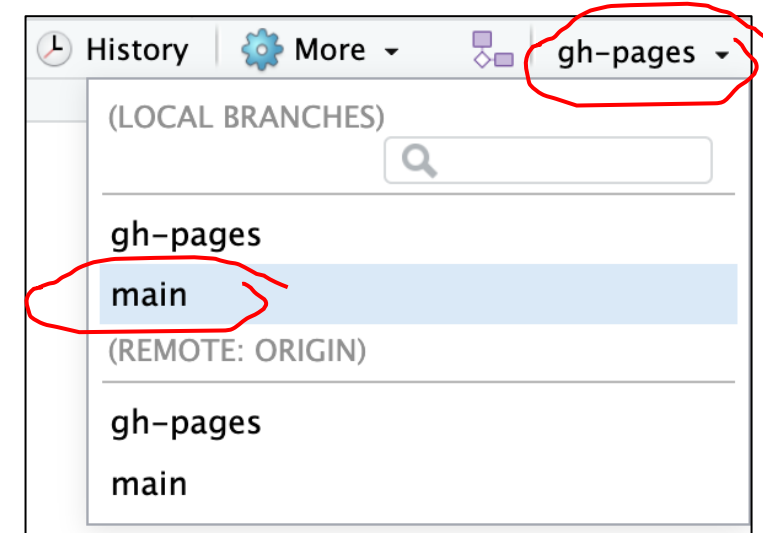
Last step: create a URL for browser to INTERPRET the HTML

3a. Create a special place (branch) for your Interpretable code

- In RStudio > Git > click 'new branch symbol' 
- Create a new branch called **gh-pages**. We'll need it in a minute.
- Then come **back to the main branch**. All of our work will happen on main.



Then...





Last step: create a URL for browser to INTERPRET the HTML

3b. Add instructions on how to publish your page (we wrote this one for you)

- In RStudio create a folder '.github' and in it folder 'workflows'
- In workflows create a file called publish.yml
- Copy-paste this code into your yml file



```
#note: only once, before first time repo is deployed, create a branch called: gh-pages.

on:
  workflow_dispatch:
  push:
    branches: main

name: Quarto Publish

jobs:
  build-deploy:
    runs-on: ubuntu-latest
    permissions:
      contents: write
    steps:
      - name: Check out repository
        uses: actions/checkout@v4

      - name: Setup R
        uses: r-lib/actions/setup-r@v2
        with:
          r-version: '4.4.0'

      - name: Setup Quarto
        uses: quarto-dev/quarto-actions/setup@v2

      - uses: r-lib/actions/setup-r-dependencies@v2
        with:
          cache-version: 4 #increase this number if github actions require some old packages - it means 'needing' them
          is remembered in cache
          packages: |
            any::markdown
            any::pak
            # any::knitr
            # any::quarto

      - name: Render and Publish
        uses: quarto-dev/quarto-actions/publish@v2
        with:
          target: gh-pages
        env:
          GITHUB_TOKEN: ${ secrets.GITHUB_TOKEN }
```



Last step: create a URL for browser to INTERPRET the HTML

3b. Add instructions on how to publish your page (we wrote this one for you)

- In RStudio create a folder '.github' and in it folder 'workflows'
- In workflows create a file called publish.yml
- Copy-paste this code into your yml file



You do not need to understand yml. Here are core parts:

on: push: branches: main

Do all this whenever someone does PUSH to main branch

name: Render with: target: gh-pages

Render everything to gh-pages branch you just created

```
#nbc: only once, before first time repo is deployed, create a branch called: gh-pages.

on:
  workflow_dispatch:
  push:
    branches: main

name: Quarto Publish

jobs:
  build-deploy:
    runs-on: ubuntu-latest
    permissions:
      contents: write
    steps:
      - name: Check out repository
        uses: actions/checkout@v4

      - name: Setup R
        uses: r-lib/actions/setup-r@v2
        with:
          r-version: '4.4.0'

      - name: Setup Quarto
        uses: quarto-dev/quarto-actions/setup@v2

      - uses: r-lib/actions/setup-r-dependencies@v2
        with:
          cache-version: 4 #increase this number if github actions require some old packages - it means 'needing' them
          is remembered in cache
          packages: |
            any::markdown
            any::pak
            # any::knitr
            # any::quarto

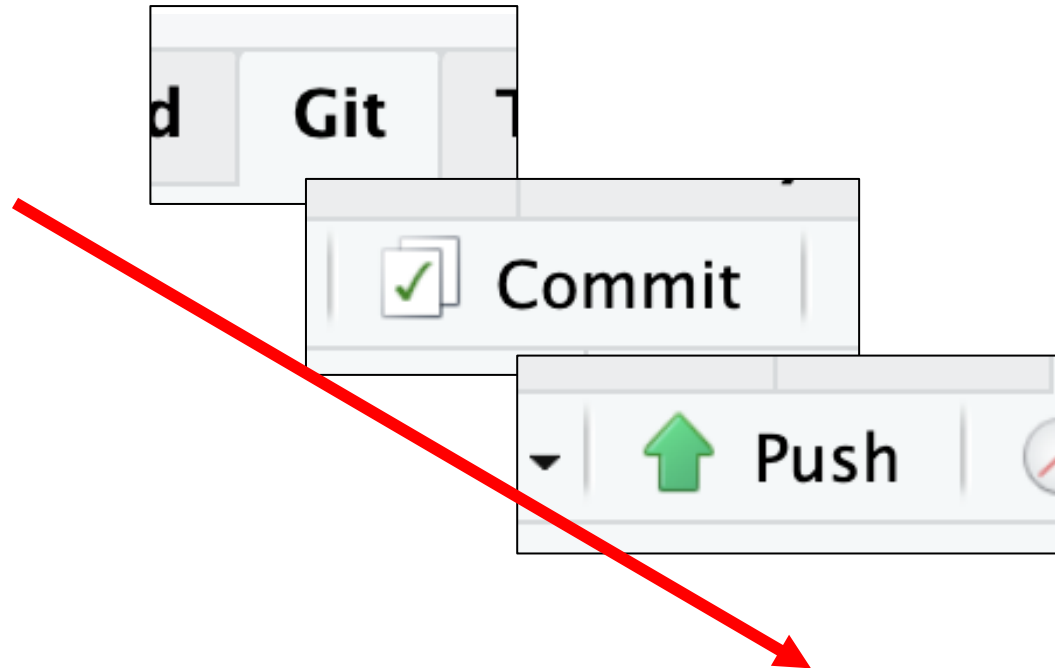
      - name: Render and Publish
        uses: quarto-dev/quarto-actions/publish@v2
        with:
          target: gh-pages
        env:
          GITHUB_TOKEN: ${ secrets.GITHUB_TOKEN }
```



Last step: create a URL for browser to INTERPRET the HTML

3c. Tell the git we made changes

In RStudio follow the git process: **Commit and Push** (green arrow) your change

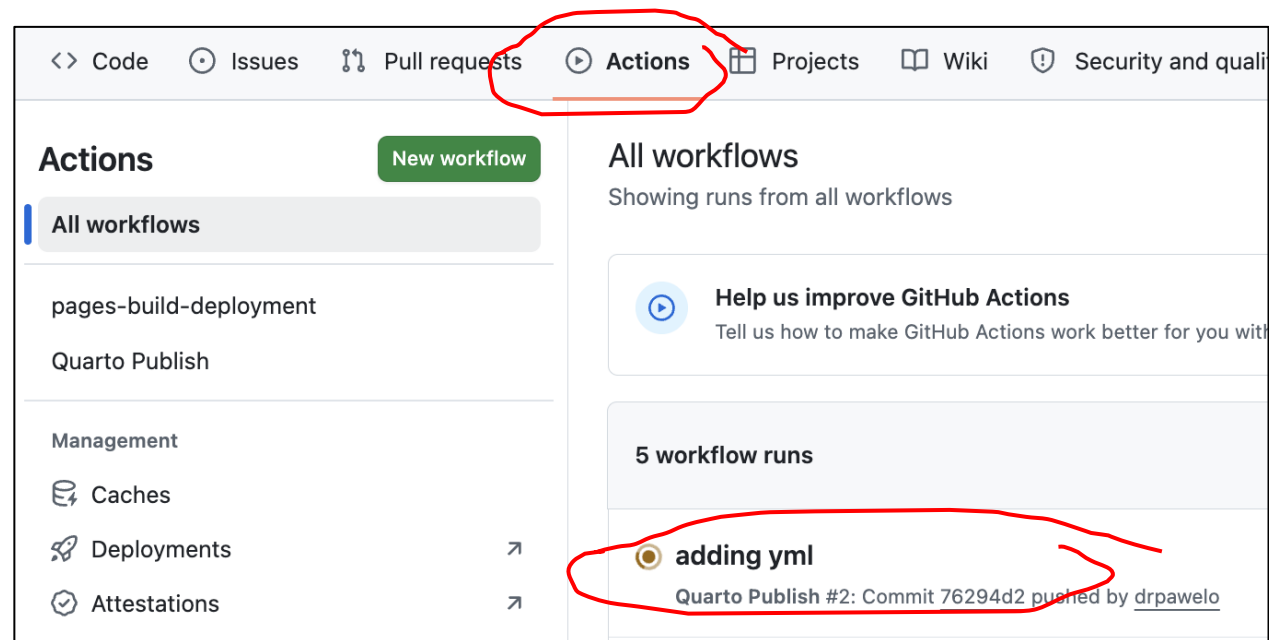




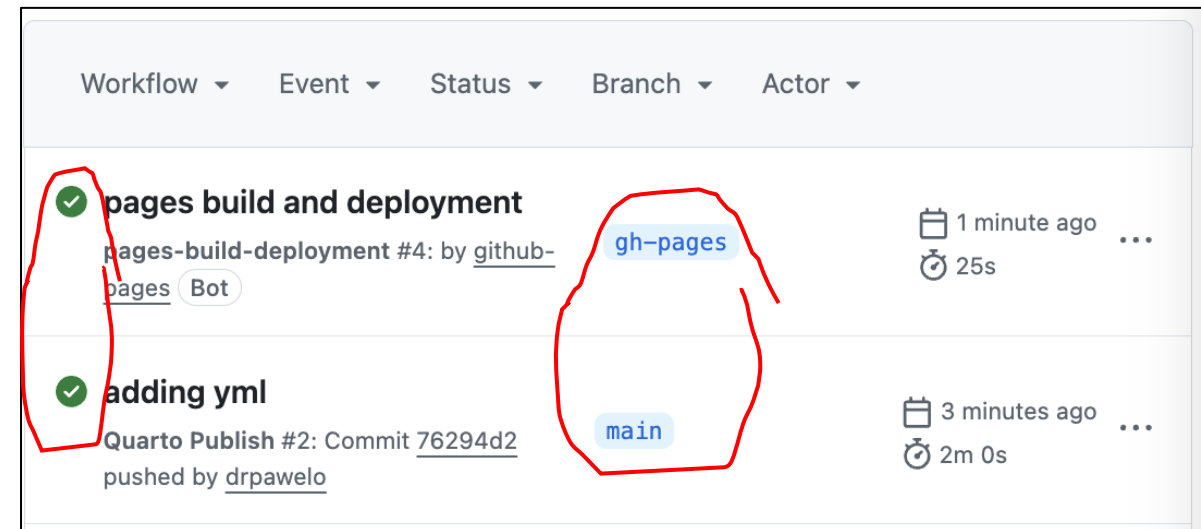
Last step: create a URL for browser to INTERPRET the HTML

3d. When we Committed and Pushed your change of the the publish.yml file

- When you Pushed, look on the Github page in Actions section
- You will see a spinning wheel, it take 2-3 minutes to create and publish your page



After 2-3 minutes, you will see both rendering and publishing worked....





Last step: create a URL for browser to INTERPRET the HTML

3e. Find the web address where your page was deployed.

- On GitHub page of your repo go to Settings > Pages > and you will see the URL

drpawelo / workshop_quarto_website_6

Code Issues Pull requests Actions Projects Wiki Security and quality Insights **Settings**

General

Access

Collaborators

Moderation

Code, planning, and automation

Rulesets

Branches

Tags

Actions

Webhooks

Copilot

Planning

Environments

Codespaces

Pages

GitHub Pages

GitHub Pages is designed to host your personal, organization, or project pages from a GitHub repository.

Your site is live at https://drpawelo.github.io/workshop_quarto_website_6/
Last deployed by [github-pages\[bot\]](#) 4 minutes ago

[Visit site](#) [Unpublish site](#)

Build and deployment

Source

Deploy from a branch

Branch

Your GitHub Pages site is currently being built from the gh-pages branch. [Learn more about configuring the publishing source for your site.](#)

[gh-pages](#) [/ \(root\)](#) [Save](#)

Learn how to [add a Jekyll theme](#) to your site.

Your site was last deployed to the [github-pages](#) environment by the [pages build and deployment](#) workflow.

Click that link to see it live!!!

workshop_quarto_website_6

workshop_quarto_website_6

This is a Quarto website.

To learn more about Quarto websites visit <https://quarto.org/docs/websites>.

```
1 + 1
```

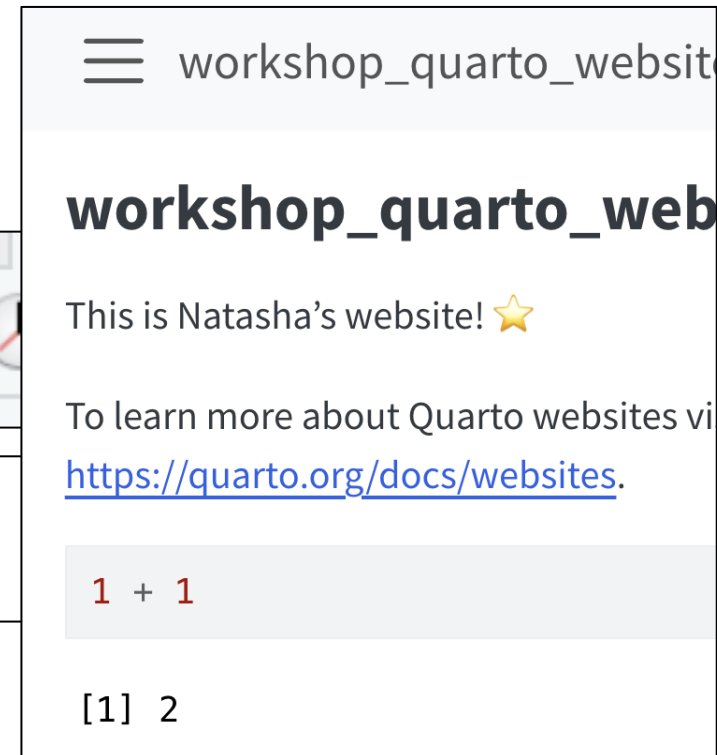
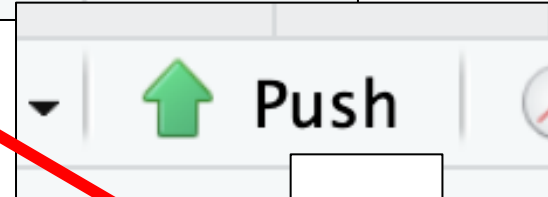
```
[1] 2
```



Last step: create a URL for browser to INTERPRET the HTML

3e. That's it. Whenever you push any changes to your repo, they will appear on your live website (after a few minutes)

```
1 ---
2 title: "workshop_quarto_website_6"
3 image: profile.jpg
4 ---
5
6 This is Natasha's website! ★
7
8 To learn more about Quarto websites visit
9 <https://quarto.org/docs/websites>.
10
11 ```{r}
12 1 + 1
13 ```
```

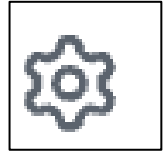




Last step: create a URL for browser to INTERPRET the HTML

3f. (optional top tip) Save your public url in repository description

- In the Code section of github page of your repo, edit repo description
- Paste the URL there, so you won't be looking for it next time

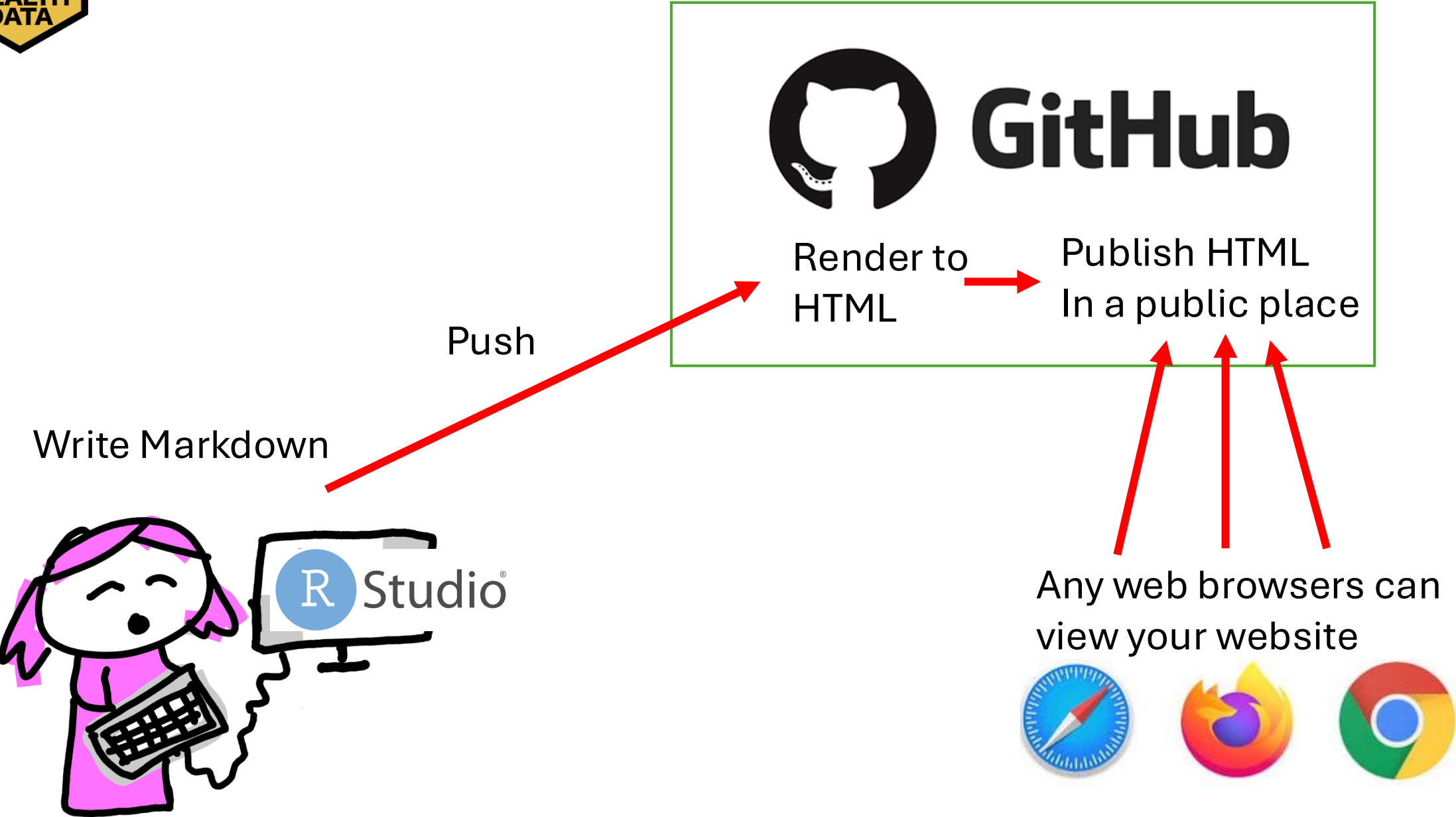


A screenshot of a GitHub repository page for "drpawelo / drpawelo-test_website_workshop_4". The "Code" tab is selected and highlighted with a red circle. The repository is public. The "About" section on the right has a gear icon circled in red. A modal window titled "Edit repository details" is open, showing the "Description" field with the URL "https://drpawelo.github.io/workshop_quarto_website_6/" pasted in, which is also circled in red. The left sidebar shows the repository's file structure, including ".github/workflows", ".quarto/project-cache", and ".gitignore".

An inset showing the "About" section of the repository. It displays the URL "https://drpawelo.github.io/workshop_quarto_website_6/" as the description, with a gear icon in the top right corner.

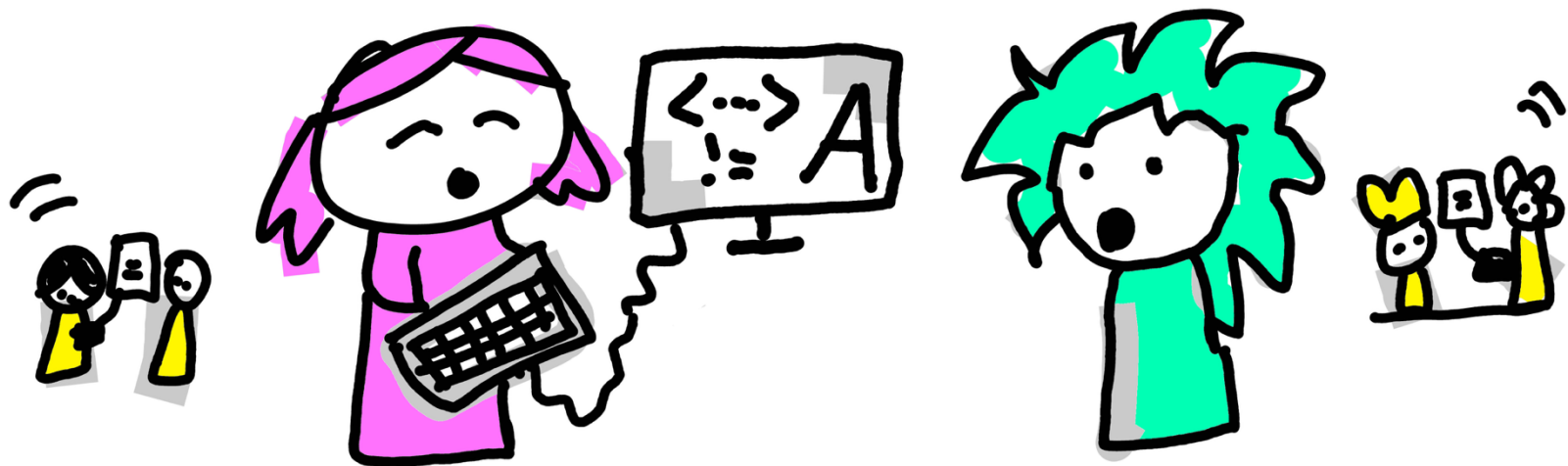


You're DONE (well.. Getting started)





**Now in Pairs go through this process for one of you,
then for the other one**





End of workshop 1



Workshop 2:

Website structure: Menus, sub-pages, blog posts, lists